

University of Michigan-Dearborn Syllabus

CIS 450 / ECE 478: Operating Systems, 4 Credits

Semester and Year: Winter 2026



Prof. Probir Roy

Office Location: CIS 230

Phone Number: (313) 583-6620

E-Mail: probirr@umich.edu

Office hours: Tue,Thu 2:00PM- 4:00PM i n office (If using zoom, please make an appointment early)

Course Meeting Times and Format(s): Tue,Thu 4:00PM-5:45PM, Lecture in Social Science

Building (SSB) 2195

Course Description This course presents the functions of an operating system as a manager of resources, including file systems, disk and storage, CPU and memory. The concepts of process and thread, synchronization mechanisms, scheduling strategies and deadlock detection/avoidance are covered in detail, along with an introduction to protection and security and distributed systems. Prerequisites: CIS 310 and (CIS 350/3501 or IMSE 350 or (ECE 370 and ECE/MATH 276)) and previous or concurrent enrollment i n IMSE 317.

Program Goals To provide students with the necessary skills to design and implement operating systems through a good understanding of three pieces of operating systems: virtualization (CPU and memory management), concurrency (multi-thread, semaphore, deadlock), and persistence (file system). It comes with hands-on machine problems such as system call design, CPU scheduling hacking, and concurrency control.

Course Objectives

- Knowledge of general principals and concepts of operating systems, in particular, the three pieces of operating systems: virtualization, concurrency, and persistence.
- Ability to design some basic operating system components.

Required Materials and/or Technology

- [Required textbook] Operating Systems: Three Easy Pieces, R. Arpaci-Dusseau and A. Arpaci-Dusseau, <http://www.OSTEP.org>. It's online and free!
- [Recommended] C Programming Online Tutorial, <https://www.tutorialspoint.com/cprogramming> For students with deficient background in C programming.
- [Recommended] xv6: a simple, Unix-like teaching operating system, Cox, Kaashoek, Morris, <http://pdos.csail.mit.edu/6.828/2018/xv6.html>. (For course projects).

University Attendance Policy Each in-classroom student is expected to attend every class. Each student needs to inform the instructor of the reason for each absence. Each distance learning student is expected to watch every recorded lecture instead. The instructor is entitled to give a failing grade (E) for excessive absences or an Unofficial Drop (UE) for a student who stops attending or watching recorded lectures at some point during the semester.

Academic Integrity The University of Michigan-Dearborn values academic honesty and integrity. Each student has a responsibility to understand, accept, and comply with the University's standards of academic conduct as set forth by the Code of Academic Conduct (<http://umdearborn.edu/697817>), as well as policies established by each college. **Cheating, collusion, misconduct, fabrication, and plagiarism** are considered serious offenses, and may be monitored using tools including but not limited to **Turnitin Simcheck**. Violations can result in penalties up to and including expulsion from the University. At the instructor's discretion, the penalty may be a grade of zero on the assignment up to and including recommending that the student be expelled from the University. It is the sole responsibility of the student to understand and follow academic guidelines regarding plagiarism. The University of Michigan-Dearborn has an online academic integrity tutorial that can be accessed at <https://webapps.umd.umich.edu/aim>.

Practice of this class: ZERO Tolerance of Violation! For any suspect of a violation, the instructor will meet with the student. For a minor violation, half credit of the whole assignment will be deducted. Otherwise, the instructor will check with College Undergraduate Advising Office to see if the student has prior violation in the university-level database. For a first time offender case, the penalty will be a grade of zero on the assignment up to. For a repeat offender case, it goes to the Code of Conduct Committee (Integrity Committee, as is referred to in the campus document) for further deliberation.

Grading Policy

- Items: This course includes (subject to change)

Items	Counts	Credits
Participation	multiple	10%
Homework	3	15%
Programming	4	40%
Exams	2	35% (Midterm 15%, Final 20%)

- Grading Scale: No curve will be used. The final grade will strictly follow this scale:

A+	100%+ – 96.67%	A	96.66% – 93.34%	A-	93.33% – 90.00%
B+	89.99% – 86.67%	B	86.66% – 83.34%	B-	83.33% – 80.00%
C+	79.99% – 76.67%	C	76.66% – 73.34%	C-	73.33% – 70.00%
D+	69.99% – 66.67%	D	66.66% – 63.34%	D-	63.33% – 60.00%
E	59.99% – 00.00%				

- Participation: Quiz will be posted on Canvas. Each student is given a 1 day to work on each quiz. Late submission is acceptable but with **1%-credit-per-hour penalty**.

- Homework assignments: All should be done **individually**. All will be due at the time specified and should be submitted through Canvas website. Late submission is acceptable but with **1%-credit-per-hour penalty**. Exceptions to these rules will be made only under exceptional circumstances, and then only with an appropriate written excuse. Many questions are similar to the ones in exams. So, make sure to finish them well.

Programming assignments: All could be done by a team with at most two members. Late submission is acceptable but with **1%-credit-per-hour penalty**.

- Exams: No late submission is allowed. Make-up exams will NOT be given unless there is prior notification of and arrangement with the instructor. In case of emergency on the day of exam, a police report, birth certificate, doctor's note or the equivalent is required for the make-up to be granted. We expect that the exams will have some bonus points.

Disability Statement The University will make reasonable accommodations for persons with documented disabilities. Students need to register with Counseling & Disability Services (DS) every semester they are enrolled. DS is located in 2157 UC (<https://umdearborn.edu/students/disability-services>). To be assured of having services when they are needed, students should register no later than the end of the add/drop deadline of each term. If you have a disability that necessitates an accommodation or adjustment to the academic requirements stated in this syllabus, you must register with DS as described above and notify the instructor.

Harassment, Sexual Violence, Bias, and Discrimination The University of Michigan-Dearborn recognizes that students have a right to study in a safe atmosphere free of sexual violence, harassment, bias and discrimination. Should you wish to report an incident of sexual assault, harassment, bias and discrimination, visit <https://umdearborn.edu/offices/enrollment-management-student-life/incident-and-complaint-reporting>.

Course schedule Check online on Canvas (subject to change).